

Extensions: mixed responses, spatial and temporal ordination

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Outline

- ▶ Mixed response types: modelling communities with different measurement scales
- ▶ Spatial and temporal autocorrelation in latent variables
- ▶ Case study: joint modelling across survey methods with spatial and temporal structure

Questions so far?



Not all species are measured the same way

Community data rarely comes in a single format:

- ▶ Species composition: presence-absence
- ▶ Dominant grasses: % cover
- ▶ Rare forbs: count of individuals
- ▶ Invertebrates: biomass

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- ▶ Invertebrates: biomass

Standard GLLVMs fit one distribution to all species. But forcing binary, count, and continuous responses into the same family means either:

- ▶ **Information loss** (discretising continuous data)
- ▶ **Model misspecification** (Poisson for proportion data)

Mixed response GLLVMs

In `gllvm`, the `family` argument accepts a **character vector** of length p (one per species column):

```
family_vec <- c(rep("binomial", 12), rep("poisson", 8),
               rep("tweedie", 4))

fit <- gllvm(y = Y, family = family_vec, num.lv = 2)
```

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```

Each species gets its own link function and variance structure. The latent variables $\mathbf{u}_i$ are shared across all species.

What distributions are available?

Family	Data type	Link
"gaussian"	Continuous, symmetric	identity
"gamma"	Positive continuous	log
"exponential"	Positive continuous	log
"tweedie"	Non-negative, right-skewed (biomass)	log
"beta"	Proportions in (0, 1), no exact 0/1	logit/probit
"orderedBeta"	Proportions including 0 and 1	logit/probit
"betaH"	Proportions including 0, excluding 1 (hurdle)	logit/probit
"poisson"	Counts	log
"negative.binomial"	Overdispersed counts (quadratic var.)	log
"negative.binomial1"	Overdispersed counts (linear var.)	log
"ZIP", "ZINB"	Zero-inflated counts	log
"binomial"	Presence-absence, or k of N trials	probit/logit/cloglog
"ZIB", "ZNIB"	Zero-/ N -inflated trials	probit/logit/cloglog
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family restricts links to what the estimation method (VA/EVA/LA) supports for that family; ?gl1vm has the full method-by-link breakdown.

Binomial with $N > 1$ trials

When species are recorded as frequencies (e.g., Raunkiaer: k quadrats occupied out of $N = 100$):

```
fit <- gllvm(y = Y, family = "binomial", Ntrials = 100,  
            num.lv = 2)
```

$Ntrials$ can also be a vector of length p if trials differ per species, or a matrix of size $n \times p$.

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`Ntrials` can also be a vector of length p if trials differ per species, or a matrix of size $n \times p$.

`binomial` supports probit, logit, or cloglog links; the default in `gllvm` is **probit**, not logit.

orderedBeta for proportions

Proportions including exact 0 and 1 (e.g., Braun-Blanquet cover converted to $[0, 1]$):

```
fit <- gllvm(y = Y / 100, family = "orderedBeta", num.lv = 2)
```

The orderedBeta distribution places point masses at 0 and 1 with a Beta distribution on $(0, 1)$. More appropriate than "beta" when true absences and full cover both occur.

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zetacutoff controls the quantile threshold for the zero/one mixture; default is usually fine, but set explicitly for stability:

```
fit <- gllvm(..., family = "orderedBeta",  
             zetacutoff = c(-2, 2))
```

Combining families in a single model

A realistic community dataset might need:

```
is_cover <- grepl("_cover$", colnames(Y))
is_count <- grepl("_count$", colnames(Y))
family_vec <- ifelse(is_cover, "orderedBeta",
                     ifelse(is_count, "negative.binomial",
                           "binomial"))

Ntrials <- ifelse(grepl("_freq$", colnames(Y)), 100, 1)

fit <- gllvm(y = Y, family = family_vec, Ntrials = Ntrials,
             num.lv = 2, zetacutoff = c(-2, 2))
```

A core assumption we have not questioned

Recall Thursday's concurrent ordination: $\mathbf{u}_i = \mathbf{B}^\top \mathbf{x}_i + \epsilon_i$. Whether unconstrained, constrained, or concurrent, every ordination so far has assumed the residual part is **independent** across sites:

$$\epsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$$

For unconstrained ordination $\mathbf{u}_i = \epsilon_i$, so this is the same iid assumption directly on $\mathbf{u}_i$.

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This fails when:

- ▶ Plots are close in space and share unmeasured environmental variation
- ▶ The same plots are revisited over time
- ▶ The design has nested structure (transects within sites)

Structured covariance for the latent variables

Thursday generalized the **mean** of $\mathbf{u}_i$ via $\mathbf{B}^\top \mathbf{x}_i$. Now we generalize the **covariance** of ϵ_i instead, replacing independent draws with a **correlated** distribution:

$$\epsilon_{.q} \sim \mathcal{N}(\mathbf{0}, \Sigma_q), \quad q = 1, \dots, d$$

where Σ_q encodes the dependence structure.

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In `gllvm`: argument `lvCor` specifies the covariance model;
`studyDesign` provides the grouping or ordering variable.

Structures available in glvm

Formula syntax	Structure	Use case
<code>~corMatern(1 site)</code>	Matérn kernel	Spatial, general
<code>~corExp(1 site)</code>	Exponential kernel	Matérn, $\nu = 1/2$
<code>~corAR1(1 time)</code>	AR(1)	Equally-spaced time
<code>~corCS(1 group)</code>	Compound symmetry	Nested designs

`corExp` is not a separate model: it is `corMatern` with smoothness ν fixed at $1/2$. More on this shortly.

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The same structures are available for row effects via `row.eff`.

You have seen `lvCor` before for nested designs (`corCS`); the same argument accepts spatial and temporal kernels.

Temporal autocorrelation: discrete time

Equally-spaced sampling (e.g. annual census): composition at time t is correlated with time $t - 1$ via AR(1):

$$\Sigma_q[t, t'] = \rho^{|t-t'|}, \quad \rho \in (-1, 1)$$

```
fit_ar1 <- gllvm(  
  y           = Y,  
  X           = Xenv,  
  formula     = ~ temperature + precipitation,  
  num.lv      = 2,  
  studyDesign = data.frame(site = site_id, year = year_id),  
  row.eff     = ~corAR1(1|year),  
  lvCor       = ~corAR1(1|year),  
  family      = "negative.binomial"
```

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The continuous-time analogue of AR(1) is the Ornstein-Uhlenbeck process:

$$\Sigma_q[t, t'] = \exp\left(-\frac{|t - t'|}{\rho}\right)$$

using the actual elapsed time $|t - t'|$, not an integer lag. This is the **same exponential kernel** used for space, applied along the time axis instead.

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Evaluated at equally-spaced integer times, this kernel reduces exactly to AR(1): $\rho_{\text{AR1}} = \exp(-\Delta t / \rho)$.

Fitting continuous-time correlation in gllvm

```
fit_ct <- gllvm(  
  y          = Y,  
  X          = Xenv,  
  num.lv     = 2,  
  distLV     = matrix(sample_time, ncol = 1),  
  lvCor      = ~corExp(1|time),  
  studyDesign = data.frame(time = time_id),  
  family     = "negative.binomial"  
)
```

The Matérn covariance function

One kernel family for space *and* time; corExp is the special case $\nu = 1/2$. Of its three parameters, gllvm **estimates** ρ , lets you **fix** ν , and fixes $\sigma^2 = 1$ (the LV's own scale is the separate sigma.lv you know from Thursday):

$$K(d) = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} d}{\rho} \right)^\nu \mathcal{K}_\nu \left(\frac{\sqrt{2\nu} d}{\rho} \right)$$

- ▶ d : distance (or elapsed time) between two points
- ▶ σ^2 : marginal variance, $K(0) = \sigma^2$ (fixed to 1)
- ▶ $\rho > 0$: range parameter, decay speed (estimated: rho.lv)
- ▶ $\nu > 0$: smoothness parameter (fixed by you: MaternKappa)
- ▶ $\mathcal{K}_\nu$: modified Bessel function, second kind, order ν

Matérn special cases

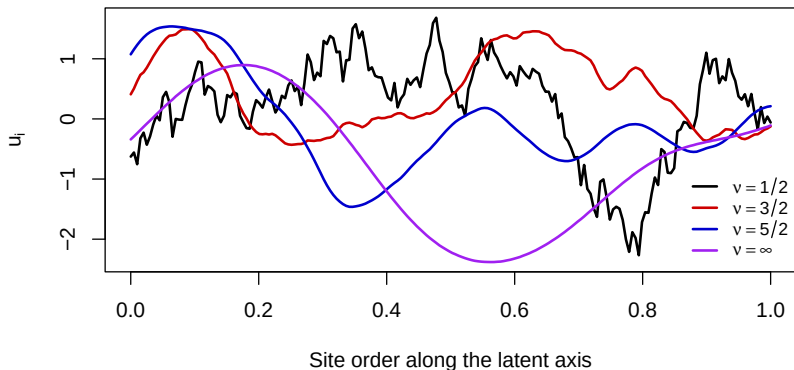
- ▶ $\nu = 1/2$: exponential kernel, $K(d) = \sigma^2 \exp(-d/\rho)$, what we used for continuous time and corExp: a rough, jagged process
- ▶ $\nu \rightarrow \infty$: Gaussian (squared-exponential) kernel, infinitely smooth
- ▶ Larger $\nu \Rightarrow$ smoother realizations

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- ▶ Larger $\nu \Rightarrow$ smoother realizations

Fixing ν (whether via corExp or MaternKappa) is not just a computational shortcut: it is a specific hypothesis about how smooth the underlying spatial/temporal environmental process is. In practice ν is very hard to estimate from data, so gllvm fixes it rather than estimating it.

Matérn paths



Four simulated realisations of the **same** latent variable and range ρ , only ν differs: this is what “fixing ν ” commits you to.

```
fit_sp <- gllvm(
  y          = Y,
  num.lv     = 2,
  distLV     = site_coords,      # n x 2 coordinate matrix
  lvCor      = ~corMatern(1|site),
  MaternKappa = 1.5,
  studyDesign = data.frame(site = site_id),
  family     = "orderedBeta"
)

fit_sp$params$rho.lv    # range in same units as site_coords
```

Interpreting the estimated range

ρ is estimated. Effective range (correlation < 0.05): $d \approx 3\rho$ (exact for $\nu = 1/2$).

Where does the correlation belong?

Row effects (row.eff)

Community-level variation:
affects all species equally at a
site.

Good for: site productivity,
observer effects, year-level
abundance shifts.

LV correlation (lvCor)

Species composition variation:
different species respond
differently to the latent
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Both can be included simultaneously:

```
fit <- gllvm(...,  
  row.eff = ~corAR1(1|year),  
  lvCor    = ~corExp(1|site))
```

Computational considerations

Two separate matrices get expensive for large n :

1. The **true** correlation matrix Σ_q itself (from the Matérn/exponential kernel) is dense: evaluating and inverting it during optimisation still does not scale well. This is a genuinely unsolved problem in `gllvm`, in the same spirit as what INLA's sparse GMRF/SPDE approximation solves for spatial models. No drop-in fix yet; keep n in mind when you reach for `lvCor`.

Sparse approximations for the other matrix

2. The **variational** covariance matrix (the approximation to the posterior) is what `gllvm` already addresses with **sparse approximations**:
 - ▶ `Lambda.struc = "bdNN"`: block-diagonal nearest-neighbour Cholesky
 - ▶ `Lambda.struc = "UNN"`: Kronecker product with NN sparse Cholesky

NN controls the number of neighbours; more neighbours = better approximation, slower fitting. For large datasets ($n > 500$), start with `NN = 5` and increase if needed.

Spatial, temporal, and phylogenetic confounding

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Consequences: biased covariate effects, distorted between-species correlations, incorrect variance partitioning, and shifted ordination axes. Their fix, **restricted spatial factor analysis**, constrains the spatial factors to be uncorrelated with the covariates.

Confounding is not specific to space

The same problem shows up wherever a structured latent variable and a covariate can vary along the same axis:

- ▶ **Temporal:** a covariate that trends over time, confounded with corAR1/corExp latent variables
- ▶ **Phylogenetic:** a phylogenetically structured trait, confounded with a phylogenetic random effect (colMat), which is one reason a phylogenetic signal estimate can vanish even when phylogeny really does matter (see Wednesday)

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Before trusting a fixed effect, check whether the covariate or trait is itself structured along the same axis as a random effect in the model.

Concurrent ordination with covariates

Yesterday we saw **concurrent ordination**: LVs *informed* by predictors, $\mathbf{u}_i = \mathbf{B}^\top \mathbf{x}_i + \epsilon_i$, not constrained to them; the residual ϵ_i keeps all variation the covariates do not explain.

With `num.lv.c` and `lv.formula`, we can combine this with spatial/temporal structure.

Fitting a concurrent ordination with spatial structure

```
fit_conc <- gllvm(  
  y          = Y,  
  X          = X,  
  family     = "poisson",  
  num.lv.c   = 2,  
  lv.formula = ~ temperature + moisture,  
  distLV     = site_coords,  
  lvCor      = ~corExp(1|site),  
  studyDesign = data.frame(site = site_id),  
  randomB    = "LV"  
)
```

The LVs are simultaneously informed by covariates and spatially correlated.

Families used in the Garching case study

Family	Data type	Link
"ordinal"	Ordered categories (Londo / BB codes)	cumulative probit
"ZNIB"	Zero-/ N -inflated frequency (k of N trials)	probit

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`zeta.struc = "common"`: cutpoints shared across species.
 Species-specific cutpoints ("`species`") are unidentifiable under a block-diagonal NA structure.

The Garchinger Heide dataset

A calcareous grassland in Bavaria (Bauer & Albrecht 2020; data: PANGAEA).

Two long-term monitoring programmes, never combined before:

Coverage survey

42 plots $\times$ {2003, 2018}
Londo ordinal codes (12 levels)
→ ordinal

Frequency survey

40 plots $\times$ {1984, 1993, 2018}
Raunkiaer (/100 quadrats)
→ ZNIB, $N = 100$

Question: do the two methods recover the same community gradient?

The joint model design

Stack both surveys as rows; shared species get two columns:

			Bromus_COV	Bromus_FREQ	Molinia_FREQ
COV, 2003	plot S03,	5		NA	NA
COV, 2018	plot S03,	7		NA	NA
FREQ, 1984	plot II5,	NA		73	12
FREQ, 2018	plot II5,	NA		81	8

- ▶ 204 rows, 196 columns; 79 shared species $\times$ 2 + survey-only
- ▶ Block-diagonal NA structure: no plot is in both surveys

What the joint model estimates

The model estimates **one set of latent variables** covering all rows, and **separate loadings** for `_COV` and `_FREQ` versions of each shared species.

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If the two survey methods agree, $\hat{\gamma}_{j,\text{COV}} \approx \hat{\gamma}_{j,\text{FREQ}}$; testable with Procrustes.

The identification problem

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If both surveys reflect the same community gradient, the separate loadings should still end up similar once the shared latent variables are estimated. But rotation ambiguity means we cannot guarantee the two blocks land in the same orientation at all.

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If both surveys reflect the same community gradient, the separate loadings should still end up similar once the shared latent variables are estimated. But rotation ambiguity means we cannot guarantee the two blocks land in the same orientation at all.

We need an external anchor.

Model 1: shared species only

```
fit3 <- gllvm(  
  y           = Y_joint,  
  family      = family_vec,  
  Ntrials     = Ntrials,  
  num.lv      = 2,  
  zeta.struc  = "common",  
  row.eff     = ~(1|site),  
  studyDesign = data.frame(site = physical_site)  
)
```

Diagnosing Model 1's alignment

Diagnose alignment with a Procrustes rotation between the shared species' _COV and _FREQ loading matrices.

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Diagnose alignment with a Procrustes rotation between the shared species' `_COV` and `_FREQ` loading matrices.

A low Procrustes correlation here signals real rotation ambiguity, motivating the spatial anchor below.

Model 2: spatial anchor

4 coverage–frequency plot pairs lie within 6 m of each other. The spatial covariance provides a partial anchor via nearby plots.

```
fit4 <- update(fit3,  
  distLV      = coords_sites,  
  lvCor       = ~corExp(1|site),  
  studyDesign = data.frame(site = physical_site),  
  Lambda.struc = "UNN")
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  Lambda.struc = "UNN")
```

Lambda.struc = "UNN": sparse nearest-neighbour Cholesky approximation;
see computational considerations slide.

Results: spatial model

Procrustes comparison of coverage and frequency loadings:

- ▶ Correlation ≈ 0.2 , protest $p = 0.081$
- ▶ Most shared species align well

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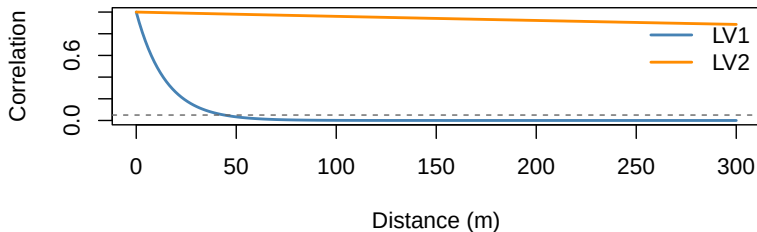
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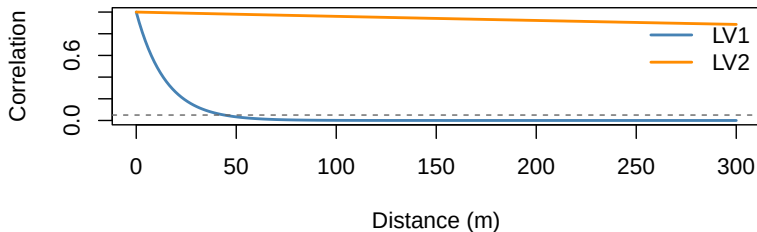
Fitted spatial range: $\hat{\rho}_{LV1} \approx 15$ m, $\hat{\rho}_{LV2} \approx 2481$ m.

Effective range ($\text{Cor} < 0.05$): ~ 44 m and ~ 7432 m.

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Most inter-plot distances exceed 80 m; the community gradient is driven by local habitat conditions, not broad spatial autocorrelation. The few close pairs provide just enough anchor.

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- ▶ Erica herbacea: calcicole specialist, sensitive to management changes
- ▶ Molinia caerulea: competitive graminoid, potentially expanding since the 1980s

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- ▶ Erica herbacea: calcicole specialist, sensitive to management changes
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Disagreement is concentrated in species that have **changed between the survey periods** (1984–1993 vs. 2003–2018), not random noise.

This is a resurvey: disagreement = ecological change, not methodological failure.

Model 3: adding temporal structure

lv.formula = ~ Year now drives the ordination axes (concurrent ordination); lvCor continues to structure only the residual ϵ_i , not the whole ordination as it would for unconstrained LVs, so site-level spatial correlation persists on top of the Year effect. Site random intercepts carry over from fit4:

```
fit5 <- update(fit4,  
  num.lv      = 0,  
  X           = X_joint,  
  num.lv.c    = 2,  
  lv.formula  = ~ Year,  
  randomB     = "LV",  
  lvCor       = ~corExp(1|site),  
  Lambda.struc = "UNN")
```

What does each component do?

- ▶ `lv.formula = ~ Year`: LVs are driven by survey year; temporal community change is captured in the ordination axes
- ▶ `row.eff = ~(1|site)`: site-level random intercepts absorb overall abundance differences across plots
- ▶ `lvCor = ~corExp(1|site)`: residual spatial structure in community composition after accounting for Year

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The Procrustes alignment between the `_COV` and `_FREQ` loadings then answers: after partialling out temporal change (via Year), do the two surveys still agree on the community gradient?

Three model perspectives

Model		Anchor	Alignment	Interpretation
fit3:	shared species	None (separate γ per survey)	Procrustes r , low	Rotation ambiguity
fit4:	+ Spatial	Nearby cross-survey pairs	$r \approx 0.2$	Methods agree spatially
fit5:	+ Year	Space + time	$r \approx 0.04$	Residual agreement

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fit4: + Spatial	Nearby cross-survey pairs	$r \approx 0.2$	Methods agree spatially
fit5: + Year	Space + time	$r \approx 0.04$	Residual agreement

The three models are not competitors: they answer **different questions** about the same data.

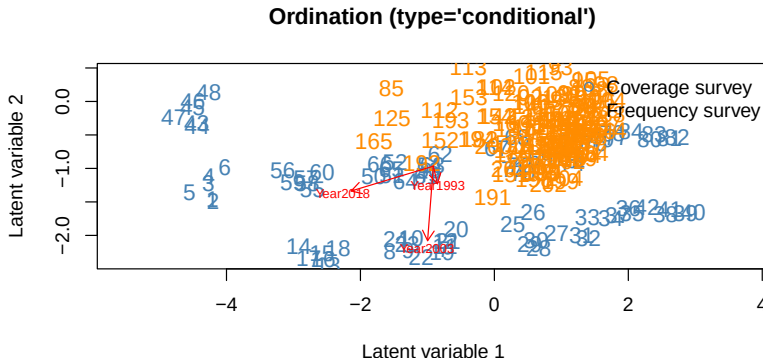
Three model perspectives

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The three models are not competitors: they answer **different questions** about the same data.

The joint family GLLVM makes all three possible within a single framework.

The joint ordination



Both surveys' plots land in the same ordination space; a common community gradient underlies both measurement methods.

Summary

Mixed responses

- ▶ family accepts a vector; each species gets its own distribution and link
- ▶ Shared LVs connect all species into one ordination space
- ▶ Enables joint modelling across datasets with different measurement scales

Spatial and temporal autocorrelation

- ▶ `lvCor` and `row.eff` accept structured covariance models (`corExp`, `corAR1`, ...)
- ▶ `distLV` provides coordinates for spatial kernels
- ▶ Short fitted ranges are informative, not a failure

Combining both

- ▶ Mixed responses + spatial LVs + concurrent ordination: all composable in `gllvm`
- ▶ The Garchinger example shows how far this can be pushed on real data